

CROWN INSTITUTE

FROM POSSIBILITY TO ACTUALITY

ICT913 - Networking

LECTURE 10

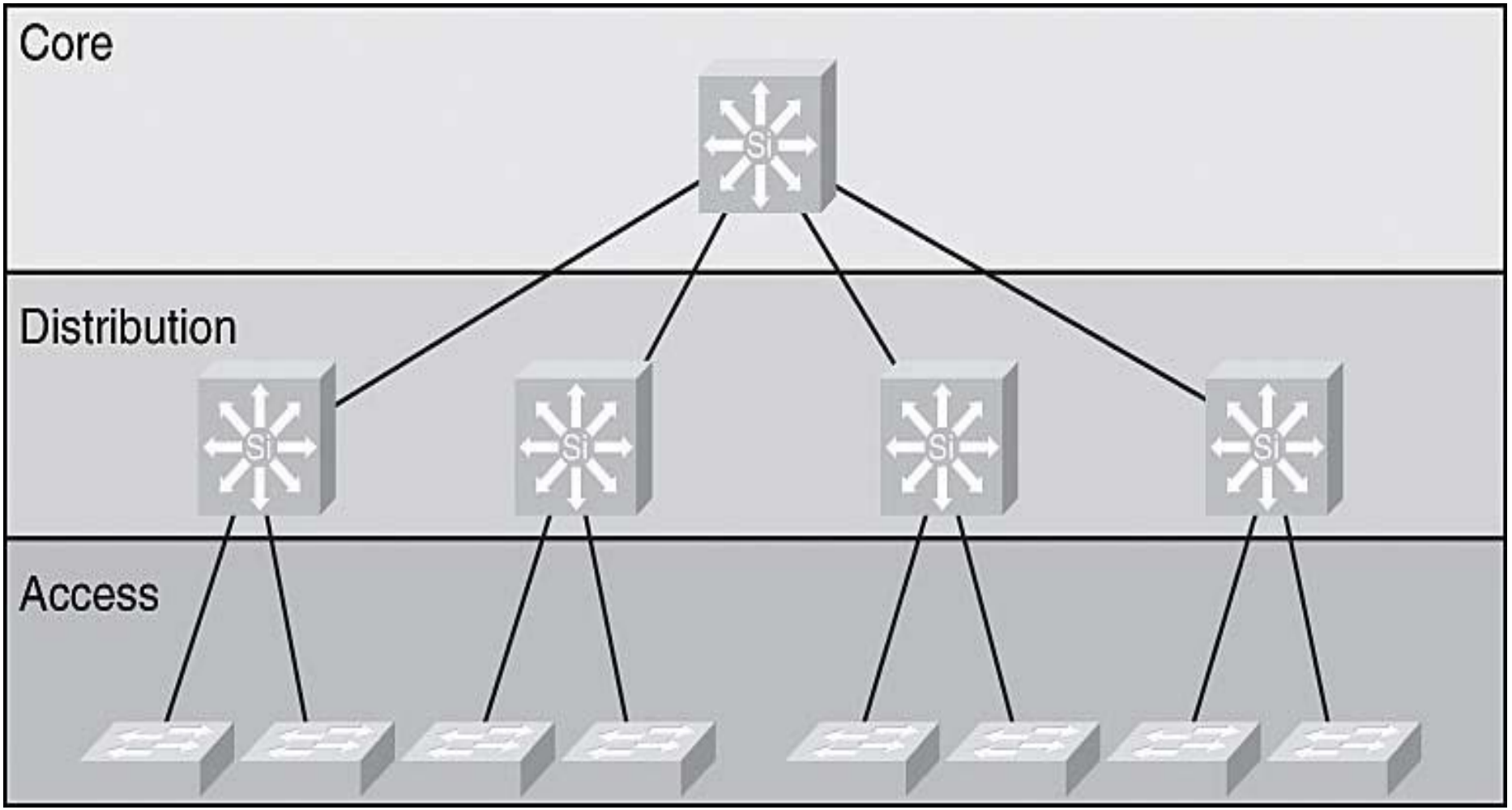
Designing Enterprise Networks

Business Constraints

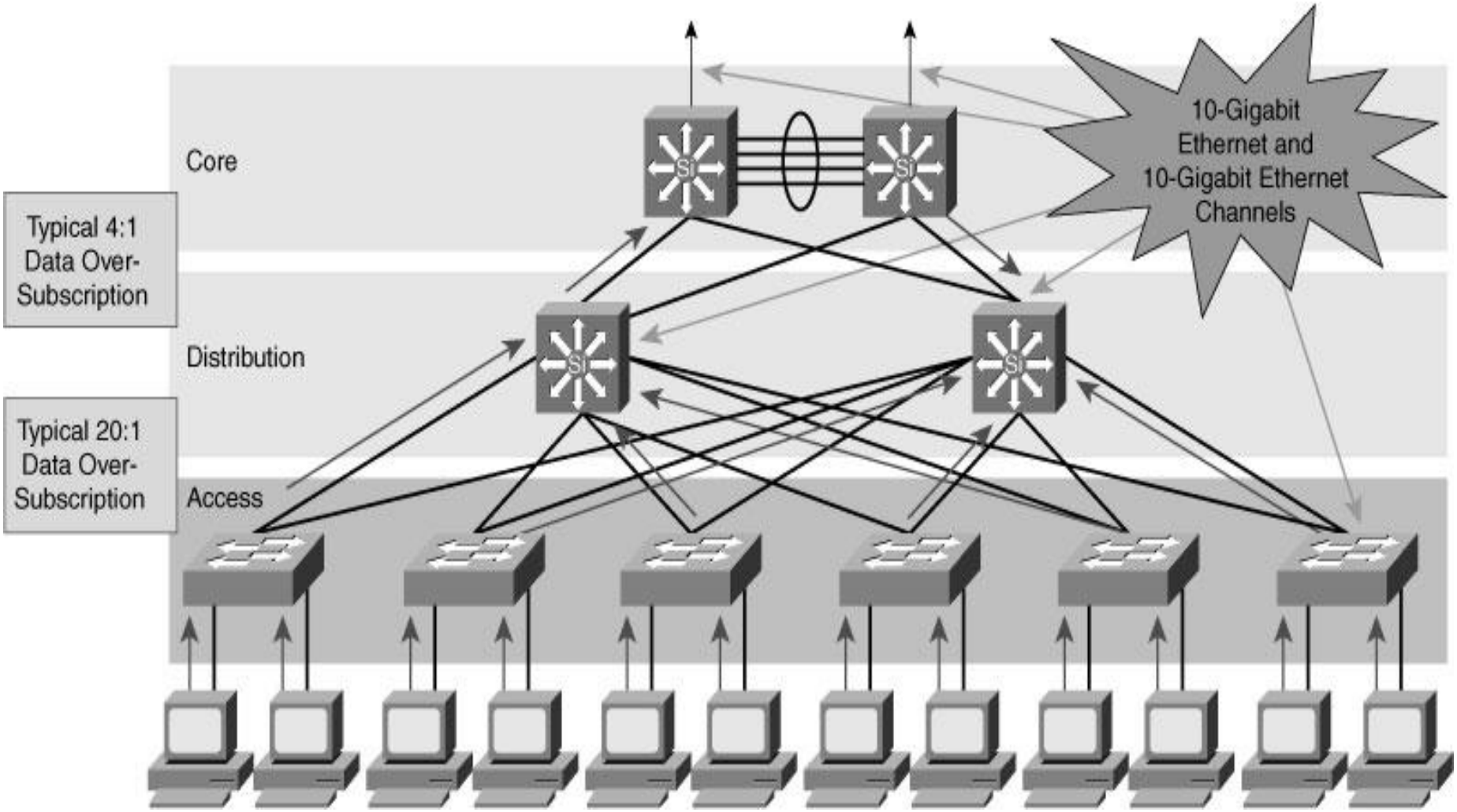
- Budget
- Staffing
- Schedule
- Politics and policies



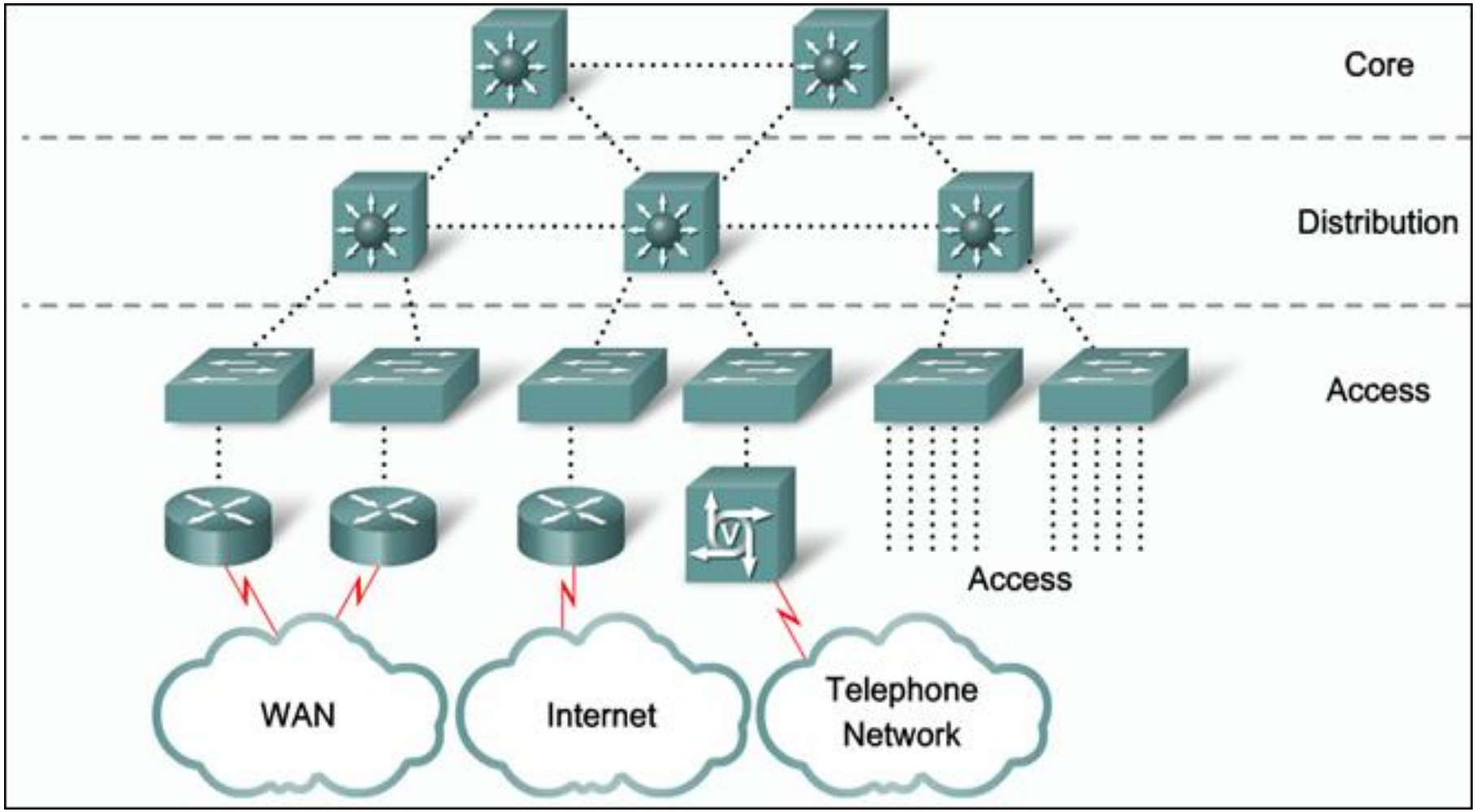
Cisco Enterprise 3 Layer Design



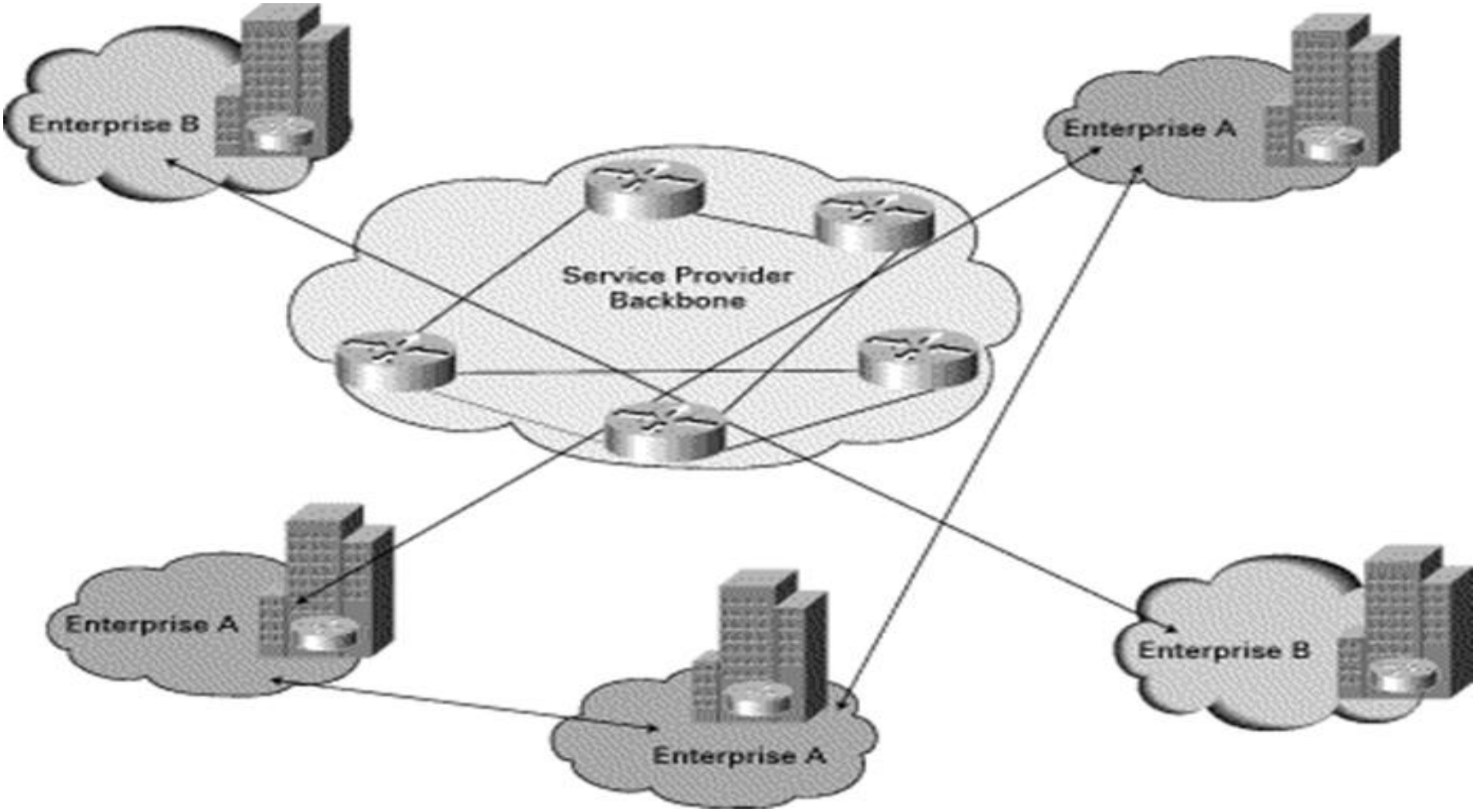
Cisco Enterprise 3 Layer Design



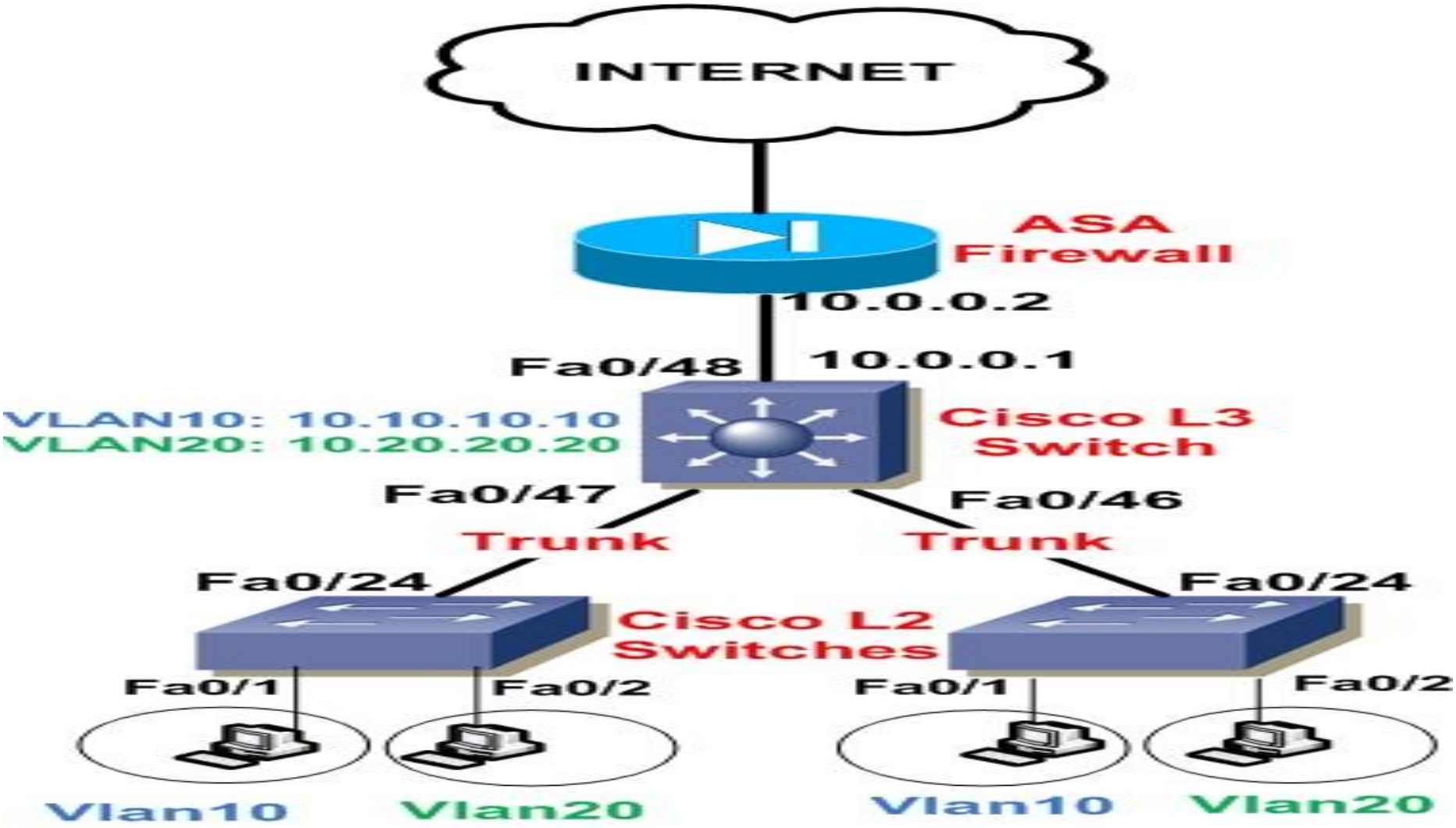
Cisco Enterprise 3 Layer Design



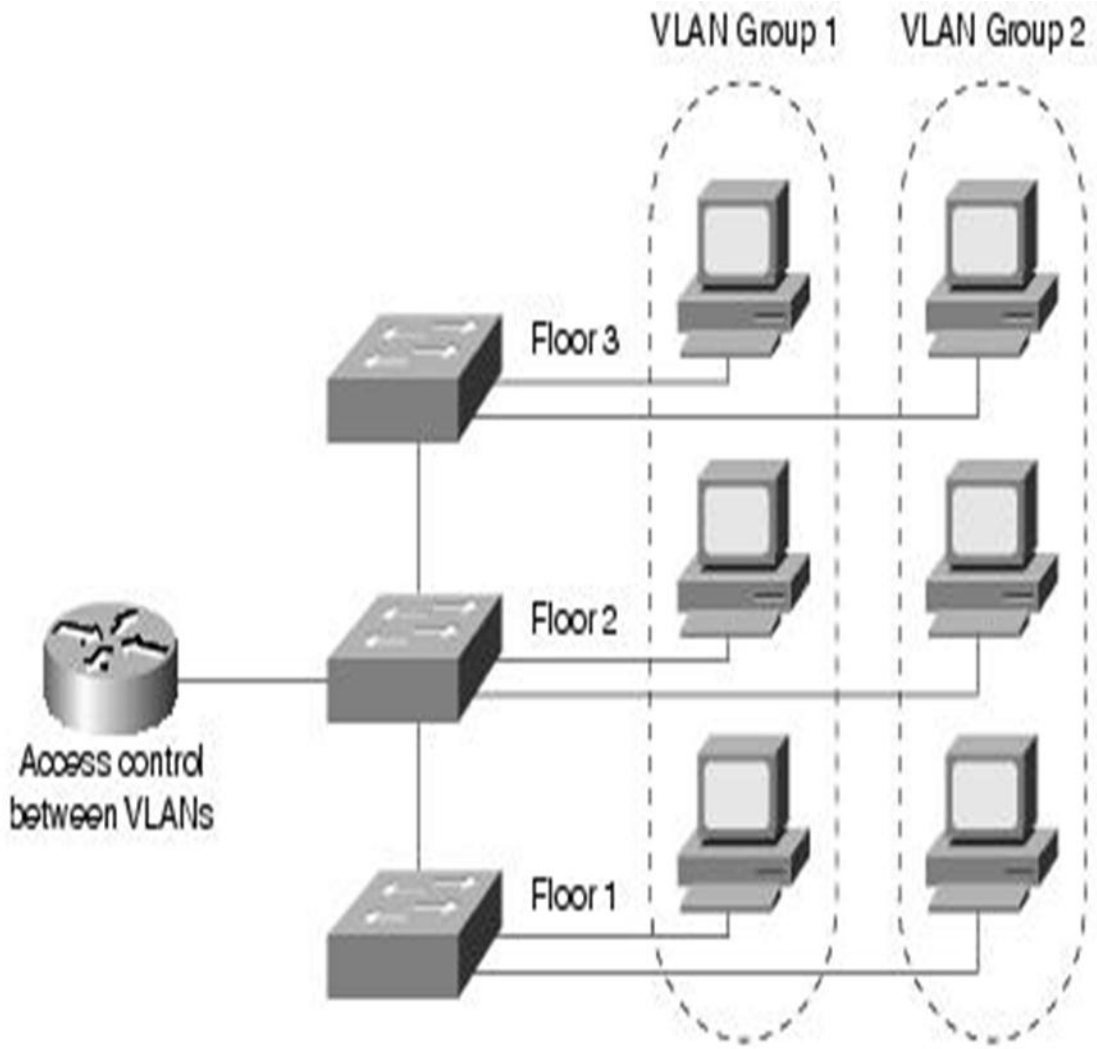
Cisco Enterprise 3 Layer Design



Cisco Enterprise 3 Layer Design



Cisco Enterprise 3 Layer Design



Network Technical Goals

- Scalability
- Availability
- Performance
- Security
- Manageability
- Usability
- Adaptability
- Affordability

Scalability

- Scalability refers to the ability to grow a network with existing hardware and software.
- How much growth is anticipated within the next 5 years?

Large companies expand more rapidly (users, applications, external networks, and new sites) than smaller ones.

- Expanding Access to Data

1970 -1980 data stored on mainframes

1980 – 1990 data stored on servers

1990 – present data stored on centralized mainframes and servers

Scalability

- 80/20 Rule

  80 percent local use and 20 percent external use. At the present time, the 80/20 Rule is moving to the other side of the scale. There is more external Internet access by employees on a daily basis (20/80).

  Some companies allow access with other companies, resellers, suppliers, and strategic customers. Introduction of the extranet.

Extranet is used to describe an internal internetwork that is accessible by outside users.

Scalability

The business goal of making data available to more departments, employees, and off site offices often results in a technical goal of using the mainframe as a powerful database server.

- Some technologies are more scalable than others.
  For example: Flat network designs at Layer 2 switches, and do not don't scale well.

  Top-down network design is an iterative process. Scalability goals and solutions are re-evaluated on a regular basis throughout the phases of the network design process.

Scalability Constraints

Constraints often affect scalability inherent in network technologies.

Selecting technologies that meet the customers scalability goals is a difficult process, especially if it is done without planning, could result in a costly re-design process later down the road.

Scalability

- Extract the information from your customer about their site. Both current and future network information.
    - Number of sites to be added in the next 5 years
    - What functionality will be needed at each of these sites?
    - How many users will be added in the next 5 years?
    - How many more servers will be added to a server farm or individual departments?

Availability

- Availability can be expressed as a percent of uptime per year, month, week, day, or hour, compared to the total time in that period.

  For example:

  - 24/7 operation
  - Network is up for 165 hours in the 168-hour week
  - Availability is 98.21%

Availability

- Different applications and areas of campus may require different levels of availability. Availability could be considered a critical goal for a network design customer.
- Some enterprises may want 99.999% or “Five Nines” availability

Availability

From a customers perspective, they want to know how much time the network will be operational.

Availability is linked to reliability.

- Reliability addresses a list of issues, which include accuracy, error rates, stability, and the time between failures.

Availability

- Redundancy is a solution to a goal of high availability. In this manner, redundancy means adding duplicate links or devices to a network to avoid network outages.

- Disaster Recovery

  Natural disaster – floods, fires, hurricanes, and earth quakes.

  Satellite outages – meteorite storms, collisions in space, solar flares, and system failures

Availability

Unnatural disaster – bombs, terrorist attacks, riots, or hostage situation.

Resiliency is the amount of stress a network can handle over time and how quickly the network can rebound or spring back from security breaches, natural and unnatural disasters, human error, and catastrophic software or hardware failures.

Availability

Note: Bank check clearing process after 9/11.

A main goal in the planning process would be to recognize which parts of the network are critical and must be maintained.

The disaster recovery plan should include keeping data backed up in one or more places that are unlikely to be affected by the disaster.

Secondly, the technologies affected by the disaster should be switched to another site with similar technologies.

Note: Canada's underground facility.

Availability

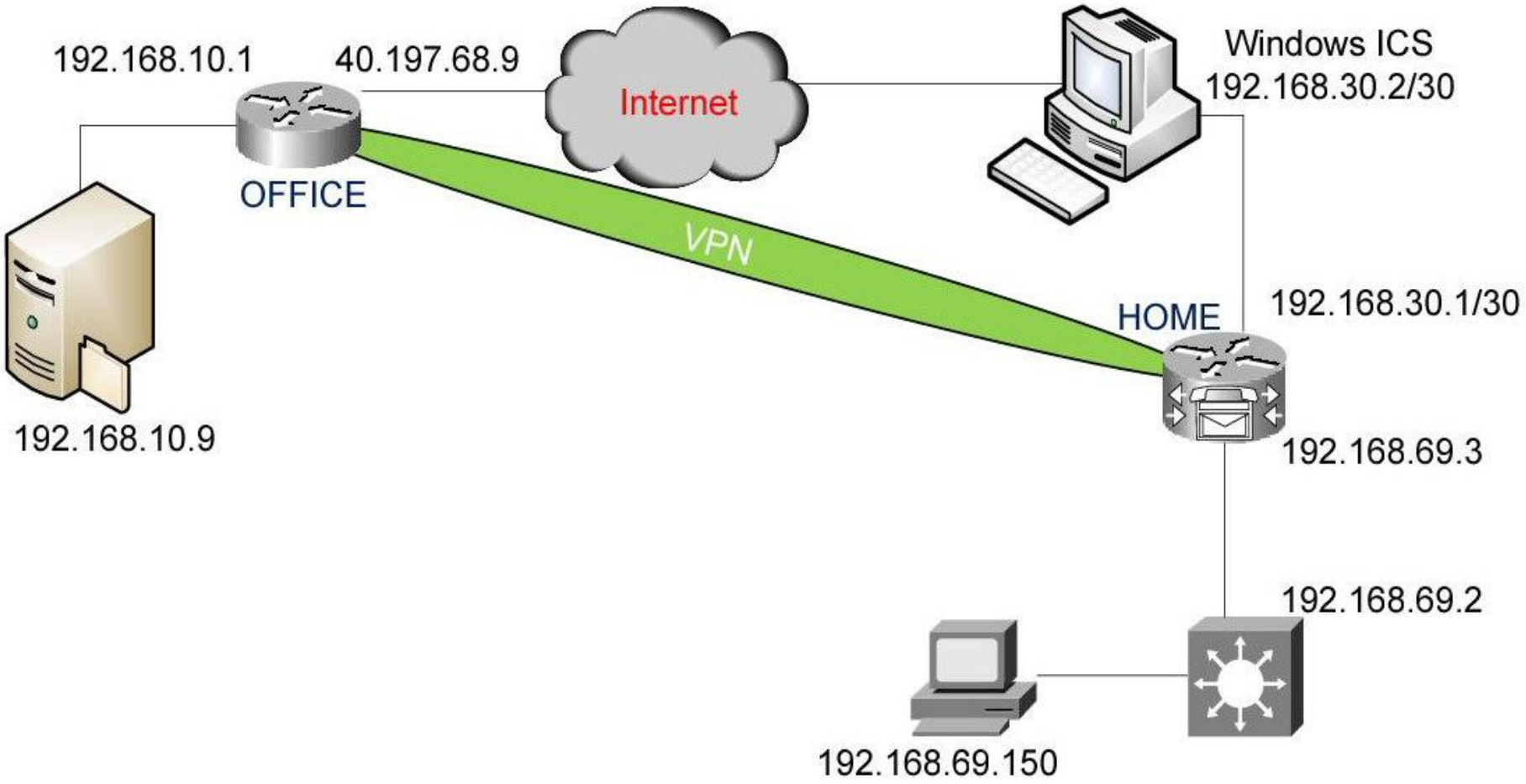
Personnel must be considered an important resource when planning for a disaster recovery.

Consider using VPN to access the corporate office when a disaster recovery occurs.

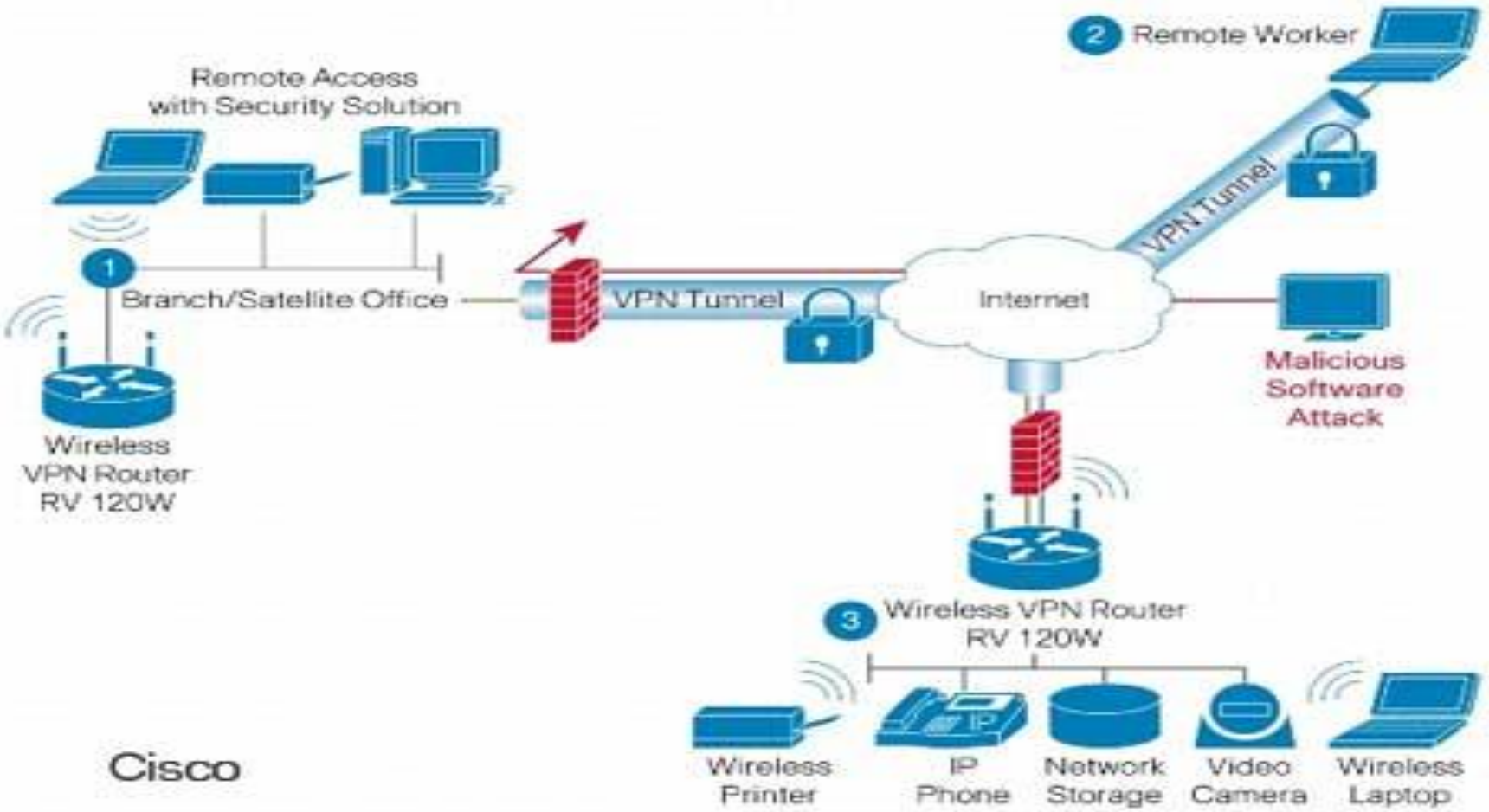
Provide VPN service to mission critical staff to work from home or a remote location.

VPN service in the case of a disaster would allow this staff to begin building the damaged system without being involved at the site where there may be contamination or disease present.

Availability



Availability



Availability

- Testing
  - It is important to require employees to be part of drills in the event of a disaster.
  - This includes visiting remote sites and utilizing the available equipment.
  - Keeping the remote equipment hardware and software at release levels similar to the main operations center.
- Availability Requirements
  - Uptime 99.95 % - network is down 5 minutes per week

Availability

- Uptime Five Nines - hard to achieve. Involves staff, equipment redundancy, and software.

  24/7 equals 8760 hours

  - Hot swappable boards
  - No maintenance window
  - In-service updates
  - Triple Redundancy
    - One active
    - One active standby
    - One standby or maintenance

http://www.youtube.com/watch?v=iKCuUX-YlMg

Availability

- Cost of Downtime
  - Each critical application should be documented. How much money the company loses per minute/hour of downtime.
  - Third party network management

Availability

- MTBF is mean time before failure
  - 4000 hours goal
- MTTR is mean time to repair
  - One hour goal

MTBF and MTTR are used to calculate available goals when the customers wants to specify explicit periods of uptime and downtime, rather than a simple percent uptime value.

Availability = (MTBF / (MTBF + MTTR))

Availability

- A typical MTBF equals 4000hours.
- A typical MTTR is 1 hour

Availability = MTBF / (MTBF + MTTR)

Availability = 4000 / (40000 + 1)

Goal 99.98 percent

- Mean times might be different in different parts of the network.
- The goal of a Cisco core layer in an enterprise network are more stringent than those goals for a switch.

Availability

- Vendors provide MTBF and MTTR estimates for their products.

- It is advisable to research independent lab results for MTBF and MTTR estimates before making a final conclusion about the product.

Network Performance

- Performance of a network includes accuracy, efficiency, delay, and response time.

- Common performance factors include
  - Bandwidth (capacity)
  - Throughput
  - Bandwidth utilization
  - Offered load
  - Accuracy
  - Efficiency
  - Delay (latency) and delay variation
  - Response time

Network Performance

- **Utilization** is normally specified as a percent of capacity.
- Optimum average network utilization is approximately 70 percent.
  - This means that peaks in the network traffic can probably be handled without noticeable performance degradation.
- Normally, WANs have less capacity than LANs.
  - When setting up the utilization estimate for a WAN links, more consideration is required regarding the bandwidths.
  - WAN links are designed with bandwidths that offer little, if any extra capacity for incidental traffic because WAN links are expensive.
- LANs are overbuilt with full-duplex Giga-bit Ethernet links to servers and 100-Mbps Giga-bit Ethernet links to clients.

Network Performance

- **Point-To-Point transmission** is a full duplex link that connects a switch to a server or some other switch.
- It is possible to consume all the bandwidth, depending on the traffic load or behavior.
- At times, network traffic appears in bursts.

http://www.youtube.com/watch?v=F32oxpCUhDM

Network Performance

- **Throughput** is the quantity of error-free data that is transmitted per unit of time.
- The assessment of the amount of data that can be transmitted per unit of time.
- **Throughput** is typically the same as **capacity**.
- Customers specify throughput goals in terms of number packets per second (PPS).
- Vendor use either PPS or cells per second (CPS) from their independent tests conducted on their product(s).
- Many internetwork devices can forward packets **a theoretical maximum**, which is called **wire speed**.

Network Performance

- **Bandwidth** means capacity and is normally fixed.
- A measure of the width of a range of frequencies.
  - Example: PVC pipe with water running through it.
- Capacity depends on the physical ISO layer.
- The capacity of a network should be adequate to handle bursts of data.

Network Performance

- **Goodput** is the number of useful bits of information at the application layer considered throughput.
- This information is delivered by the network to a certain destination, per unit of time.
- This is related to the amount of time from the first bit of the first packet is sent until the last bit of the last packet is delivered.
- **Goodput** is a measure of good and relevant application layer data transmitted per unit of time.

Network Performance

Application Layer Throughput

- Vendors refer to the **application layer throughput** as **goodput**.
- Being called **goodput**, heightens the fact that it is a measure of good and relevant application layer data transmitted per unit of time.
- **Throughput** means bytes per second (BPS).
- Applications using **throughput** as **goodput** would file transfers and data base applications.

Network Performance

- **Accuracy** is paramount when sending and receiving data.
- The data sent over the wire is expected to be identical to the data received at the destination.
- **Typical causes of data errors.**
  - Power surges or spikes
  - Impedance mismatches
  - Poor physical connections
  - Failing devices
  - Noise from electrical devices
  - Some specific software bugs

Network Performance

- **Ethernet errors** usually result from collisions.
- The error is termed, cyclic redundancy check (**CRC**).
- Errors can occur at
  - *the preamble*,
  - *past the preamble*, and
  - *beyond the 64 bytes after the preamble*.

Network Performance

- **Accuracy** refers to the number of error-free frames transmitted relative to the total number of frames transmitted.
- **Efficiency** is a measurement of how effective an operation is in comparison to the cost in effort, energy, time, and money.

  Note: *Large and small frame sizes.*

  - **Large frame** make better use of bandwidth and improve application throughput.
  - **Bigger frames** introduce more bit errors and a need for an elaborate recovery procedure.
- Response delays are expected to be minimal.
  - Variations in delay, called **jitter**

Network Performance

- **Propagation delay** is the amount of time it takes for the head of the signal to travel from the sender to the receiver (186,000 miles per second)
- **Serial delay** is the time to put digital data onto a transmission line.
- **Packet-switching delay** is the latency accrued when switches and routers forward data.
  - DRAM
  - SRAM

Network Performance

- **Queuing delay** is the time a job waits in a queue until it can be executed.
  - A good rule is to inform the customer that they should experience less than delay 1 or 2 percent
- **Response time** is the network performance goal that users are interested in.
  - Users begin to get frustrated if the response is longer than 100ms (1/10th of a second).

Security

- Focus on requirements first (MD5 / AES combined)
- Detailed security planning
- Identify network assets
  - Including their value and the expected cost associated with losing them due to a security problem.
- Analyze security risks
  - Hackers compromise a network device, such as a switch, router, server, firewall, or IDS.

Network Assets

- Hardware
- Software
- Applications
- Data
- Intellectual property
- Trade secrets
- Company’s reputation

Security Risks

- Hacked network devices
  - Data can be intercepted, analyzed, altered, or deleted
  - User passwords can be compromised
  - Device configurations can be changed
- Reconnaissance attacks
- Denial-of-service attacks
- Security should not disrupt the company's business.

Note: BOTNETS and high-capacity servers.

Manageability

Some customer goals are specific. They want to visualize problems occurring on the network. They use **SNMP** to capture the number of bytes each router receives and sends.

The FCAPS Model:

- **Fault management** – detecting, isolating, and correcting problems
- **Configuration management** – controlling, operating, identifying, and collecting data
- **Accounting management** – accounting of network usage to allocate costs to network users and/or plan for changes in capacity requirements
- **Performance management** – analyze traffic and application behavior to optimize a network, meet service-level agreements, and plan for expansion
- **Security management-** Monitoring and testing security and protection policies, maintaining passwords, encryption keys, and auditing adherence to security policies.

Usability

- **Usability**: the ease of use with which network users can access the network and services.
  - VPN might be a consideration for flexible access.
- Networks should make users' jobs easier
- Some design decisions will have a negative effect on usability:
  - **Strict security**, for example financial institutions and government offices

Adaptability

- **Avoid** incorporating any design elements that would make it hard to implement *new technologies in the future.*
- **Change** can come in the form of new protocols, new business practices, new fiscal goals, new legislation.
- **A flexible design** can adapt to changing traffic patterns and Quality of Service (QoS) requirements.

Affordability

- A network should carry the maximum amount of traffic possible for a given financial cost.
- **Affordability** is especially important in campus network designs.
- WANs are expected to cost more, but costs can be reduced with the proper use of technology
  - Quiet routing protocols, for example